

# Atif Ahmad Khan

Curriculum Vitae

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## Educational Qualifications

|             |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2026</b> | Ph.D. in Mathematics (Thesis Submitted), Aligarh Muslim University, Aligarh.<br><b>Thesis Topic:</b> A Study of algebraic aspects of Cryptography involving MDS Matrices.<br><b>Supervisor:</b> Prof. Shakir Ali, Aligarh Muslim University, Aligarh.<br><b>Co Supervisor:</b> Dr. Bhupendra Singh, Centre for Artificial Intelligence & Robotics, DRDO, Bengaluru. |
| <b>2019</b> | M.Sc. Mathematics, Aligarh Muslim University, Aligarh, CGPA: 7.710.                                                                                                                                                                                                                                                                                                 |
| <b>2017</b> | B.Sc. Mathematics, Aligarh Muslim University, Aligarh, 74.4%.                                                                                                                                                                                                                                                                                                       |
| <b>2013</b> | Intermediate, J. N. V. M. Inter college, Kanpur, 88.6%.                                                                                                                                                                                                                                                                                                             |
| <b>2011</b> | High School, J. N. V. M. Inter college, Kanpur, 80.04%.                                                                                                                                                                                                                                                                                                             |

## Academic Achievements

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|-------------|---------------------------------------------------------------------------------------------------------|
| <b>2022</b> | Qualified for the <b>CSIR–UGC Junior Research Fellowship (JRF)</b> , June 2022 with All India Rank 108. |
| <b>2021</b> | Qualified the Graduate Aptitude Test in Engineering (GATE) with an All India Rank (AIR) of 180.         |
| <b>2019</b> | Qualified the <b>CSIR–NET, December 2019</b> with All India Rank 56.                                    |
| <b>2018</b> | Qualified the <b>IIT Joint Admission Test for Masters (JAM)</b> , 2018 with All India Rank 380.         |

## Project

**M.Sc. Project:** A Study of Polynomial Rings, under the supervision of Dr. Muzibur Rahman Mozumder, Aligarh Muslim University, Aligarh, India.

## Field of Specialization/Interest

Ring Theory, Group Theory, Linear Algebra, Coding Theory, Cryptography, and Engineering Mathematics.

## Research Experience

5+ years of research experience in the fields of Cryptography and Coding Theory.

## Research Publications (Research Papers)

- Published in Refereed Journals: **4**
- Accepted for Publication: **1**

- Communicated for Publication: 4
- Papers Presented at Conferences: 5
- Conferences/ Workshops Attended: 11

## Academic Profiles

|                |                                                                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Google Scholar | <a href="https://scholar.google.com/citations?user=eHPMVh0AAAAAJ&amp;hl=en">https://scholar.google.com/citations?user=eHPMVh0AAAAAJ&amp;hl=en</a> |
| Scopus ID      | <a href="https://www.scopus.com/authid/detail.uri?authorId=59008264600">https://www.scopus.com/authid/detail.uri?authorId=59008264600</a>         |
| ResearchGate   | <a href="https://www.researchgate.net/profile/Atif-Khan-49?ev=hdr_xprf">https://www.researchgate.net/profile/Atif-Khan-49?ev=hdr_xprf</a>         |
| ORCID          | <a href="https://orcid.org/0009-0009-2371-5707">https://orcid.org/0009-0009-2371-5707</a>                                                         |
| Web of Science | <a href="https://www.webofscience.com/wos/author/record/LSL-7518-2024">https://www.webofscience.com/wos/author/record/LSL-7518-2024</a>           |

## Fellowships and Grants

- Upgraded from Junior Research Fellowship (JRF) to Senior Research Fellowship (SRF) under the CSIR-UGC Research Fellowship Scheme, 2024.
- Qualified CSIR-UGC Joint Research Fellowship (JRF), June 2022.
- Awarded financial support by **Centre International de Mathématiques Pures et Appliquées (CIMPA), France**, to attend the **CIMPA Research School 2025** at **Universitas Gadjah Mada, Indonesia**.
- Received financial support from **Centre International de Mathématiques Pures et Appliquées (CIMPA), France**, and the **Southeast Asian Mathematical Society (SEAMS)** to attend the **SEAMS School 2024** at **Universitas Diponegoro, Indonesia**.
- Awarded the **University Merit Financial Award** during the academic year 2018–2019.

## Foreign Visit

- (i) Attended the *CIMPA Research School 2025*, entitled **Arithmetic in Action: Number Theory and its Applications to Cryptography and Coding Theory**, held at *Universitas Gadjah Mada (UGM)*, Yogyakarta, Indonesia, during July 14–25, 2025.
- (ii) Attended the *SEAMS School 2024* on **Derivations on Rings and Modules with Applications**, organized by the Department of Mathematics, Faculty of Science and Mathematics, *Universitas Diponegoro*, Indonesia, during July 1–9, 2024.

## Computer Skills

- Completed the certified online course **Programming for Everybody (Getting Started with Python)** offered by the **University of Michigan** through Coursera (Verified Certificate: <https://coursera.org/verify/ZXANI6CBH5Y1>).
- Completed the certified online course **Python Data Structures** offered by the **University of Michigan** through Coursera (Verified Certificate: <https://coursera.org/verify/ZGZJGW54FXS9>)
- LaTeX, SageMath, and Microsoft Word

## Research Publications

- [1] S. Ali; [A. A. Khan](#); B. Singh: On circulant involutory and orthogonal MDS matrices over finite commutative rings. In: *Applicable Algebra in Engineering, Communication and Computing* **36**(2024), 1151-1165 (SCIE)(Q3, IF: 0.65).
- [2] S. Ali; A. S. Alali; [A. A. Khan](#); I. E. Wijayanti; K. B. Wong: XOR count and block circulant MDS matrices over finite commutative rings. In: *AIMS Mathematics* **9**(11)(2024), 30529-30547 (SCIE)(Q1, IF: 1.7).
- [3] S. Ali, N. Aydin, P. Sharma, E. S. Oztas, [A. A. Khan](#): Constacyclic codes over general mixed alphabets and their applications. In: *Designs, Codes and Cryptography* **93**(2025), 3733-3760 (SCIE)(Q2, IF: 1.2).
- [4] S. Ali; A. S. Alali; M. Azeem; [A. A. Khan](#); K. B. Wong: On double cyclic codes over finite chain rings for DNA computing. In: *Entropy* **27**(12)(2025) (SCIE)(Q2, IF: 2.0).
- [5] Defita; S. Ali; [A. A. Khan](#); I. Muchtadi-Alamsyah; A. Barra: The non-existence of  $g$ -circulant orthogonal MDS matrices over finite commutative rings. Accepted for publication in *Montes Taurus Journal of Pure and Applied Mathematics* (2026). (SCOPUS)
- [6] [A. A. Khan](#); S. Ali; B. Singh: On the generalization of  $g$ -circulant MDS matrices. In: *arXiv preprint* arXiv:2602.10028 (2026). (Submitted)
- [7] [A. A. Khan](#); S. Ali; E. S. Oztas; A. Kesarwani: MDS matrices from skew polynomials with automorphisms and derivations. In: *arXiv preprint* arXiv:2602.01383 (2026). (Submitted)
- [8] S. Ali; [A. A. Khan](#); A. Kesarwani: On the construction of Cauchy MDS matrices over Galois rings via nilpotent elements and Frobenius maps. In: *arXiv preprint* arXiv:2512.19306 (2025). (Submitted)
- [9] S. Ali; [A. A. Khan](#); A. Kesarwani; S. Samanta: Quasi-recursive MDS matrices over Galois rings. In: *arXiv preprint* arXiv:2512.17256 (2025). (Submitted)

## Papers Presented at Conferences

- 1. [International Conference on Recent Trends in Mathematics and Computing](#) (ICRTMC-2026) held at Department of Mathematics, Jamia Millia Islamia, New Delhi, INDIA, during February 16-18, 2026.
- 2. [Indian Mathematical Society Conference-International Meet](#) organized by Department of Mathematics, MIT World Peace University, Pune, INDIA, during December 23-26, 2024.
- 3. [The 1st International Conference on Mathematics and Mathematics Education](#) (ICMME 2024) organized by IndoMS DIY–Central Java, INDONESIA, during December 5–6, 2024.
- 4. [First Joint IIIMT–Algebra Forum Conference 2023](#) held at the Department of Mathematics and Natural Sciences, Mataram University, Mataram, INDONESIA, during December 16–17, 2023.
- 5. [10th International Congress on Fundamental and Applied Sciences](#) (ICFAS-2023) held at the Fatih Sultan Mehmet Vakif University, Istanbul, TURKIYE, during June 6-8, 2023.

## Conferences / Workshops Attended

1. Attended the **CIMPA Research School 2025**, entitled **Arithmetic in Action: Number Theory and its Applications to Cryptography and Coding Theory**, held at **Universitas Gadjah Mada (UGM), Yogyakarta, Indonesia**, during July 14–25, 2025.
2. Attended the SEAMS School 2024 held at **Universitas Diponegoro, Indonesia**, from July 1 to July 9 on the topic **Derivations on Rings and Modules with Applications**.
3. Research visitor at **Universitas Gadjah Mada, Indonesia** during July 11 to July 17, 2024.
4. NIWC-2023, National Instructional Workshop on Cryptology: Code-based Cryptography, during 5-7 July, 2023, **IIT BHU, INDIA**.
5. Annual Foundation School (AFS-II), Organized by **IISER Kolakata, INDIA** in 2022.
6. Academic visit at **Centre for Artificial Intelligence and Robotics, DRDO, Bengaluru, INDIA** in 2022.
7. The 23rd International Conference on Cryptology in India (INDOCRYPT 2022) organized by the **Institute for Advancing Intelligence, TCG Centres for Research and Education in Science and Technology, Kolkata, INDIA**, during December 11–14, 2022.
8. Annual Foundation School (AFS-I) in 2021.
9. Science Academic Lecture Workshop on Essence of Partial Differential Equation at **IIT Mandi, Himachal Pradesh, INDIA** in 2018.
10. Summer Programme in Mathematics at **Harish Chandra Research Institute, Prayagraj, INDIA** in 2018.
11. Fractional Derivative and its Application at Department of Mathematical Science, **IIT BHU, INDIA** in 2017.
12. Science Training Programme at **IIT Kanpur, INDIA** in 2007.

## Professional Memberships

- Life Member, Cryptology Research Society of India (CRSI).
- Membership ID: L/1132.

## Languages

- Hindi and Urdu (Native Proficiency)
- English (Professional Proficiency)

## Personal Information

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|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                      | Atif Ahmad Khan                                                                                                                         |
| <b>Father's Name</b>             | Shamim Ahmad Khan                                                                                                                       |
| <b>Mother's Name</b>             | Kahkashan Khatoon                                                                                                                       |
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| <b>Telephone</b>                 | +91-9760615215                                                                                                                          |
| <b>Date of Birth &amp; Place</b> | July 01, 1997, Ballia, Uttar Pradesh, India                                                                                             |
| <b>Marital Status</b>            | Unmarried                                                                                                                               |
| <b>Passport Number</b>           | Available                                                                                                                               |

## References

**Professor Shakir Ali**

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